

Lecture 24 Notes: Free Cash Flows, WACC, and Capital Structure

Corporate Finance

I. Non-Cash Charges and Capital Expenditures

Definition

Non-Cash Charges (NCC) are expenses subtracted on the income statement that do not involve an actual cash outflow (e.g., depreciation, amortization).

Capital Expenditures (CapEx)

- Investment in fixed assets (property, plant, equipment)
- Calculated as the change in gross fixed assets
- If an asset increases from \$10,000 to \$12,000, $\text{CapEx} = \$2,000$
- If fixed assets decrease, CapEx is negative (asset sale)

II. Net Operating Working Capital (NOWC)

Definition

Net Operating Working Capital (NOWC) represents the changes in current assets and current liabilities related to operations.

Components

- **Current Assets:** Inventory, accounts receivable
- **Current Liabilities:** Accounts payable

Change in NOWC (ΔNOWC)

$$\Delta\text{NOWC} = \Delta\text{Inventory} + \Delta\text{Receivables} - \Delta\text{Payables}$$

Key Insight

When a company expands (new plant, increased capacity), inventory and receivables typically increase. These are often financed by payables. The **net change** is what matters for free cash flow calculation.

III. Free Cash Flow to the Firm (FCFF)

FCFF Formula

$$\text{FCFF} = \text{EBIT} \times (1 - T) + \text{Depreciation} - \text{CapEx} - \Delta\text{NOWC}$$

Where:

- EBIT = Earnings Before Interest and Taxes
- T = Tax rate
- CapEx = Capital Expenditures
- ΔNOWC = Change in Net Operating Working Capital

Why Use EBIT?

FCFF represents cash available to **all capital providers** (debt holders + equity holders), so we use EBIT (before interest) rather than net income.

Non-Cash Charges

Depreciation is added back because it was subtracted to calculate EBIT but is not a cash outflow.

IV. Firm Valuation Using FCFF

Constant Growth Model

$$V_{\text{Firm}} = \frac{\text{FCFF}_1}{\text{WACC} - g}$$

Multistage Growth

For supernormal growth periods:

1. Calculate FCFF for each year during the supernormal growth period
2. Calculate the terminal value at the end of the supernormal period using constant growth
3. Discount all FCFFs and the terminal value to present using WACC

$$V_{\text{Firm}} = \sum_{t=1}^n \frac{\text{FCFF}_t}{(1 + \text{WACC})^t} + \frac{\text{Terminal Value}}{(1 + \text{WACC})^n}$$

Calculating Intrinsic Value per Share

$$V_{\text{Equity}} = V_{\text{Firm}} - \text{Market Value of Debt} - \text{Market Value of Preferred Stock} \quad (1)$$

$$\hat{P}_0 = \frac{V_{\text{Equity}}}{\text{Shares Outstanding}} \quad (2)$$

Important Note

Always use **market values** of debt and preferred stock (not book values) when calculating equity value from firm value.

V. Outstanding Shares

Definition

Outstanding shares are the shares currently held by investors in the market.

Key Distinctions

Term	Definition
Authorized Capital	Maximum shares the company can issue (by charter)
Issued Capital	Shares actually issued to investors
Outstanding Shares	Shares currently in public hands (excludes treasury shares)
Treasury Shares	Shares repurchased by the company

Why Shares Outstanding Matter

- Companies can repurchase shares (buybacks)
- Treasury shares are not counted as outstanding
- Outstanding shares = Issued shares - Treasury shares

VI. Market Value vs. Book Value Weights in WACC

Key Principle

Ideally, use **market value weights** rather than book value weights when calculating WACC.

Why Market Value Weights are Preferred

- Book values are based on historical costs
- Market values reflect current economic reality
- WACC represents the current cost of capital

Calculating Market Values

$$\text{Market Value of Equity} = \text{Shares Outstanding} \times \text{Current Market Price per Share} \quad (3)$$

$$\text{Market Value of Debt} = \text{Bonds Outstanding} \times \text{Current Market Price per Bond} \quad (4)$$

$$\text{Market Value of Preferred} = \text{Preferred Shares} \times \text{Current Market Price per Share} \quad (5)$$

Example: Effect of Market Values on Debt Ratio

- Book value of equity = \$20 million (1 million shares at \$20 book value)
- Market price per share = \$80
- Market value of equity = 1 million $\times$ \$80 = \$80 million
- Book value of debt = \$30 million
- **Book value debt ratio** = $30 / (30 + 20) = 60\%$
- **Market value debt ratio** = $30 / (30 + 80) = 27.3\%$

Key Insight

Using market values often shows firms are less leveraged than book values suggest, because equity market values typically exceed book values.

VII. The Cost of Debt Increases with Leverage

Marginal Cost of Debt

As a company takes on more debt, the **cost of debt increases** because lenders demand higher interest rates to compensate for increased default risk.

Effects of Increasing Debt

- Lenders become concerned about repayment ability
- Interest rates (marginal cost) rise at each debt level
- Shareholders also demand higher returns because financial risk increases
- The overall WACC may eventually increase with excessive debt

VIII. Understanding Financial Ratios

Debt-to-Assets Ratio

$$\text{Debt-to-Assets} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

Debt-to-Equity Ratio

$$\text{Debt-to-Equity} = \frac{\text{Total Debt}}{\text{Total Equity}}$$

Relationship Between Ratios

$$\text{If Debt-to-Equity} = 0.5 = \frac{1}{2} \quad (6)$$

$$\text{Then Total Parts} = 1 + 2 = 3 \quad (7)$$

$$\text{Debt-to-Assets} = \frac{1}{3} = 33.3\% \quad (8)$$

Conversion Formula

$$\text{Debt-to-Assets} = \frac{\text{Debt-to-Equity}}{1 + \text{Debt-to-Equity}}$$

$$\text{Debt-to-Equity} = \frac{\text{Debt-to-Assets}}{1 - \text{Debt-to-Assets}}$$

IX. Adjusting WACC for Project Risk

Key Assumption

The standard WACC represents the **average risk** of the firm. Individual projects may have different risk levels.

Risk Adjustment Rules

Project Risk Level	WACC Adjustment	Decision Rule
Above-average risk	Add a risk premium	Higher hurdle rate
Average risk	Use firm WACC	Standard hurdle rate
Below-average risk	Subtract a risk premium	Lower hurdle rate

Investment Decision

Accept project if: Project Return > Adjusted WACC

Key Insight

Firms have more margin on the **investment side** (can create competitive advantage) than on the financing side (markets are efficient).

X. Investment vs. Financing Decisions

Investment Decisions	Financing Decisions
What assets to buy	Where to get funds
Which projects to pursue	Debt vs. equity
Building competitive advantage	Bank loans, bonds, stock issuance
Firms have more control	Markets are efficient; little margin

Key Points

- Investors are smart; information is quickly incorporated into security prices
- Monetary policy changes (e.g., KIBOR) instantly affect debt costs
- Firms cannot reliably "beat" the financing market
- Competitive advantage comes from superior investment decisions

XI. The Pecking Order Theory

Definition

The **Pecking Order Theory** describes the sequence a firm follows when raising capital, prioritizing sources from least to most costly in terms of information asymmetry.

The Pecking Order

Internal Cash → Debt → New Equity

1. Internal Cash (Retained Earnings) - First Choice

- No flotation costs
- No information asymmetry
- No dilution of control
- **Condition:** Retained earnings should only be kept if the firm has investment opportunities with returns exceeding shareholders' required return.

2. Debt - Second Choice

- Interest is tax-deductible (tax shield)
- Less information asymmetry than equity
- No dilution of ownership
- Creditors are less sensitive than equity investors

3. New Equity - Last Resort

- High flotation costs
- Sends negative signal (managers may think stock is overvalued)
- Dilutes existing shareholders' control
- Equity investors demand higher returns due to residual claim

Financing Deficit

$$\text{Financing Deficit} = \text{Investment Needs} - \text{Internal Cash Flow}$$

The deficit is filled first with debt, then with new equity.

Key Insight

Return on investment must exceed the required return for retained earnings to be justified. Otherwise, funds should be returned to shareholders.

XII. Summary of Key Formulas

Formula Summary

$$\text{FCFF : } \text{FCFF} = \text{EBIT}(1 - T) + \text{Depreciation} - \text{CapEx} - \Delta\text{NOWC} \quad (9)$$

$$\text{Firm Value (Constant Growth) : } V_{\text{Firm}} = \frac{\text{FCFF}_1}{\text{WACC} - g} \quad (10)$$

$$\text{Equity Value : } V_{\text{Equity}} = V_{\text{Firm}} - V_{\text{Debt}} - V_{\text{Preferred}} \quad (11)$$

$$\text{Intrinsic Value per Share : } \hat{P}_0 = \frac{V_{\text{Equity}}}{\text{Shares Outstanding}} \quad (12)$$

$$\text{Market Value of Equity : } \text{MV}_{\text{Equity}} = \text{Shares} \times P_0 \quad (13)$$

$$\text{Debt-to-Assets from Debt-to-Equity : } D/A = \frac{D/E}{1 + D/E} \quad (14)$$

$$\text{Debt-to-Equity from Debt-to-Assets : } D/E = \frac{D/A}{1 - D/A} \quad (15)$$

XIII. Key Terms Glossary

Term	Definition
Non-Cash Charges (NCC)	Expenses that reduce net income but do not involve cash outflow
CapEx	Investment in fixed assets
NOWC	Net Operating Working Capital (current assets minus operating current liabilities)
FCFF	Free Cash Flow to the Firm (cash available to all investors)
Outstanding Shares	Shares currently held by public investors
Market Value Weights	Proportions based on current market prices
Pecking Order	Financing priority: internal cash → debt → new equity
Financing Deficit	Investment needs minus internal cash flow